



Climate change disease

A new fatal, drug-resistant disease called *Candida auris* could have emerged because of climate change. <https://dgit.in/aug19-09>



Unknown sounds mess with brains

Americans who heard some mysterious sounds in Cuba clusters of symptoms similar to a concussion. <https://dgit.in/aug19-10>

the life forms found in these cryopegs. While similar to other microbes found in marine environments, there were signs that the bacteria had evolved in isolation for over 50,000 years.

Researchers do not fully understand the process that led to the formation of the cryopegs. One theory is that the layers were from receding oceans during the last ice age. Snow fell over remaining water and turned into permafrost. The trapped water remained in a liquid state because of high salinity. The fact is that the water are the remnants of ancient oceans, and were trapped beneath the permafrost.

Studying the life trapped within cryopegs is important for future exploration of ice worlds in the Solar System. These include the moons around gas giants, many of which may have global subsurface liquid water oceans. Salty water may be trapped in temperatures below the freezing point of water, under the surface of Mars. The tortured terrain of Europa indicates that in the ice shell around the briny ocean, there may be trapped pockets of rising water, or lakes beneath the surface, and above the ocean.

There are three high profile missions to the icy moons of gas giants lined up over the next decade. These include the JUpiter ICy moons Explorer (JUICE) mission by ESA to explore Europa, Ganymede and Callisto. The Europa Clipper mission by NASA will also have a lander component. All these missions will be equipped with payloads that identify biosignatures, or signs of life. The Dragonfly mission to the Saturnian moon of Titan will study how far prebiotic chemistry has advanced on the surface. These are the confirmed missions. Apart from these, there are other concepts in the planning and conceptualisation stage, including a submarine probe to explore the global subsurface ocean on Europa, probes to study Enceladus and Titan over a series of flybys, and balloons to explore the thick atmosphere on Titan.

Among all these icy moons, perhaps the most interesting is Titan, as it can actually be hospitable to two forms of life, one as we know it on Earth, and a theoretical kind that we have not yet seen before. The hydrocarbon seas on the surface can serve the same func-



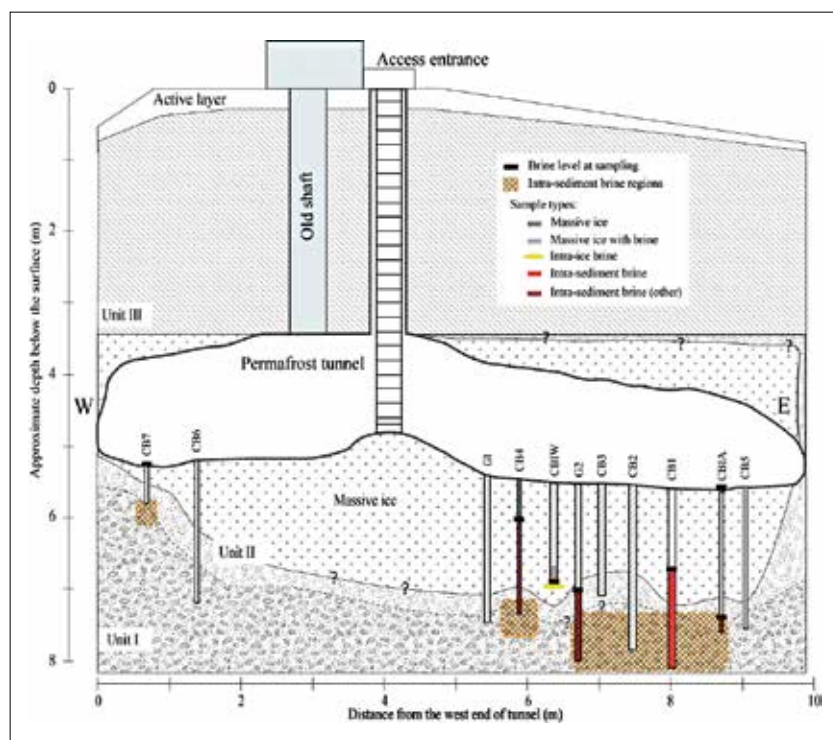
Image: NASA

Artist's concept of a subsurface lake on Europa

tion as liquid water on Earth, acting as a solvent for life forms. These forms of life could breathe in hydrogen, breathe out methane, and depend on acetylene and ethane as an energy source. If this exotic form of life exists on Titan at all, it is likely to be widespread across the surface. Beneath the surface, a more familiar form of life could exist in the subsurface oceans.

In fact, all across the galaxy, ice worlds may be the norm for finding life as against rocky planets in orbit around the habitable zone of their host stars. This is because the sheer number of ice moons are expected to exceed rocky exoplanets where liquid water can exist on the surface. The estimates for how many more vary, but there are expected to be between 100-to-1000 times more ice worlds than rocky exoplanets in the Milky Way. Given the right conditions, if these worlds have surface or subsurface oceans, they can actually be more common habitats for life than planets geologically similar to the Earth.

The finding of thriving microbial colonies in the deep brine deposits, buried beneath the Arctic permafrost and isolated from the rest of the Earth, provides a glimpse into the forms of life that could potentially exist on the icy moons in the Solar System. The studies on Earth can prepare scientists about what to look for, and what to expect, during future explorations of the ice worlds. 



A schematic of the study site